

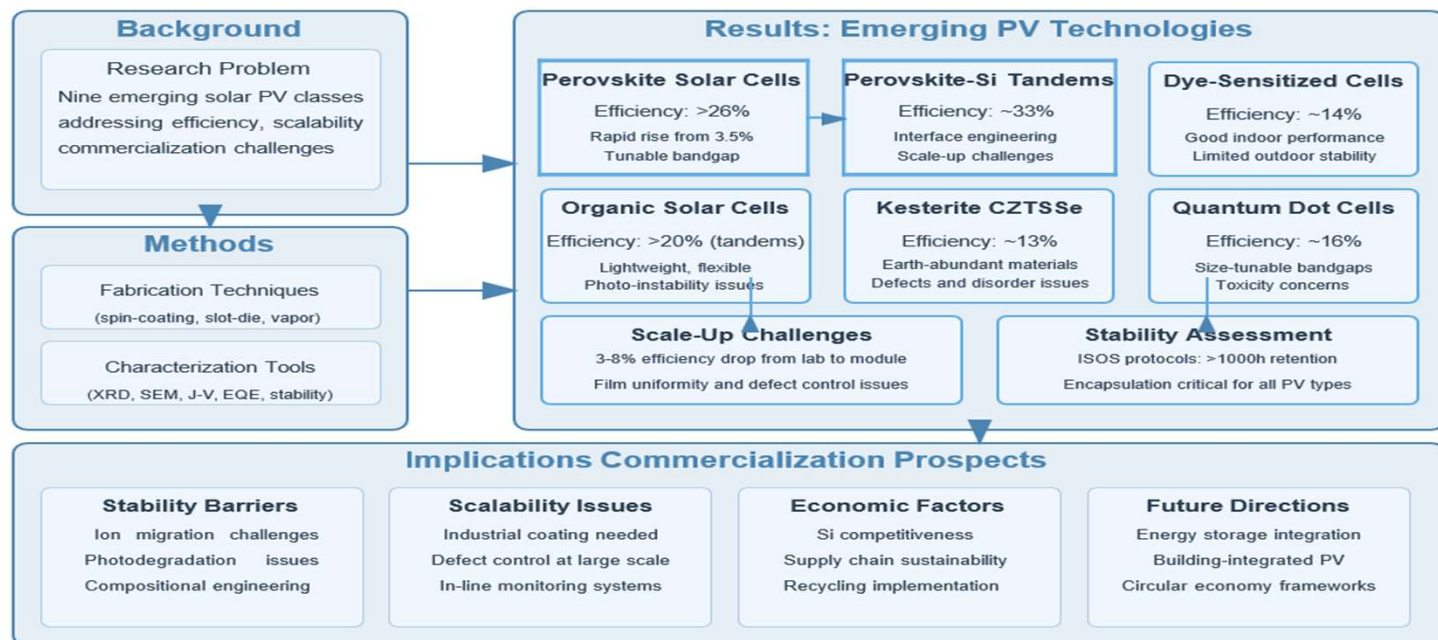


Emerging Solar Photovoltaics: From Laboratory Breakthroughs to Commercial Reality



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Introduction & Background

- Nine emerging solar photovoltaic (PV) classes are reviewed addressing efficiency, scalability, and commercialization challenges beyond silicon dominance.
- The field faces challenges in applying efficient, low-cost alternatives to crystalline silicon suitable for diverse applications.
- Advances underpinning these developments include perovskite materials, tandem cells, dye-sensitized, organic, kesterite, quantum dot, and thin absorber technologies.
- The objective is to explore emerging PV technologies as efficient, low-cost alternatives to silicon for varied applications.

Methods

- The review analyzes device physics, materials, fabrication, characterization, and scalability methods of emerging photovoltaic technologies.
- Fabrication techniques include spin-coating, slot-die, inkjet, vapor-based processes, ALD, and sequential deposition, varying by technology and scale.
- Characterization tools employed cover XRD, SEM, TEM, UV-Vis-NIR spectroscopy, photoluminescence, current-voltage (J-V), EQE measurements, and ISOS stability testing protocols.

Discussion & Implications

- The results highlight stability, scale-up, and economic considerations as principal barriers to widespread deployment of emerging PV technologies.
- Stability issues related to ion migration and photodegradation are actively mitigated by compositional engineering and effective encapsulation strategies.
- Scalability requires industrial coating processes, rigorous defect control, and real-time in-line monitoring to ensure uniform large-area module fabrication.
- Economic competitiveness with traditional silicon photovoltaics, including sustainable supply chains and recycling, is critical for market success.
- Future directions involve integration with energy storage, building-integrated photovoltaics (BIPV), and adoption of circular economy frameworks to enhance sustainability and application scope.

Results

Performance of Emerging Photovoltaic Technologies

- Perovskite solar cells reached certified efficiencies greater than 26%, improving rapidly from approximately 3.5%. They feature tunable bandgaps and solution processability but face stability and lead toxicity challenges.
- Perovskite-silicon tandem cells achieved certified efficiencies up to about 33%, with technical challenges including interface engineering, mechanical mismatch, and scale-up difficulties.
- Dye-sensitized solar cells evolved from less than 8% to around 14% efficiency, demonstrating good indoor performance though limited outdoor stability.

Additional Emerging PV Technologies and Scale-Up

- Organic solar cells and tandems attained efficiencies exceeding 19% for single junctions and 20% for tandems; they are lightweight and flexible but suffer from photo-instability restricting outdoor use.
- Kesterite ($\text{Cu}_2\text{ZnSn}(\text{S,Se})_4$) devices achieved approximately 13% efficiency using earth-abundant materials but are limited by defects and disorder.
- Quantum dot solar cells exhibited efficiencies near 16%, with size-tunable bandgaps; toxicity and stability remain concerns.
- Scale-up challenges cause efficiency drops of 3-8% when transitioning from laboratory-scale to module-scale devices due to film uniformity, defects, and interconnect losses.

Stability and Commercialization Prospects

- ISOS stability testing protocols show perovskite solar cells can retain more than 90% of their initial power conversion efficiency after 1000 hours at 55°C in nitrogen atmosphere, demonstrating the critical role of encapsulation in device stability.
- Commercialization advances are most pronounced for perovskites, with flexible, indoor organic and dye-sensitized solar cells carving niche markets; kesterite and quantum dots have roles in sustainable niche applications.

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Condensed Matter Days – 29-31 October, 2025 (CMDAYs-2025)
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